Same grade for everyone, but not everyone does the work

A UNIVERSAL PROBLEM

A meta-analysis of 78 studies (Karau & Williams, 1993) found social loafing across cultures. Unclear roles and weak individual accountability are key drivers in higher education (Piezon & Ferree, 2008).

NO SAFE WAY TO ADDRESS IT

In collectivist cultures, norms around avoiding conflict can lead active members to take over others' work, leaving passive members without consequences (Tjosvold et al., 2009). Reporting the issue to a lecturer can be seen as violating group norms.

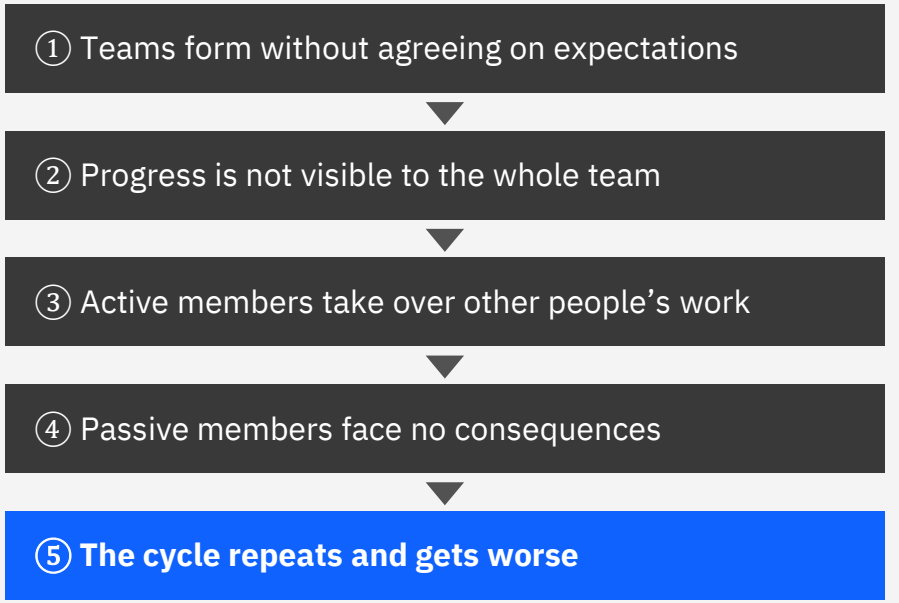
EXISTING TOOLS FALL SHORT

Trello and Notion track tasks, but they do not understand team membership or task types, and cannot detect an uneven workload.

THREE CLEAR GAPS

There is no explicit role allocation, no shared view of progress, and no accountable record of individual contributions. Without individual accountability, collective assessment enables free-riding, a pattern documented across hundreds of studies (Johnson & Johnson, 2009).

CYCLE THAT KEEPS THE PROBLEM GOING



Without intervention, this pattern persists.

Hard-working students cannot prove what they contributed. Students who are struggling have no safe way to ask for help. Both lose under the same system.

Three people in every team

Undergraduate students in Surabaya, across all majors (T1) and technical majors (T2). The critical users are passive members: without their participation, the system cannot work. This is a design assumption pending field validation.

Rahma, The Driver

De facto team coordinator

"I'm tired of being the only one chasing everyone."

PAIN POINT

There is no way to prove she does more work, or show the imbalance without triggering conflict.

KEY NEED

A documented contribution record she can show to her lecturer.

Dito, The Busy One

A student with a packed schedule

"I want to contribute, but I don't know where to start."

PAIN POINT

He does not know where to start and is afraid to ask for clarification in the group chat because he might look incompetent.

KEY NEED

Clear tasks and reminders that do not embarrass him.

Sinta, The Unsure One

A first-year student

"I don't want to look stupid asking in the group chat."

PAIN POINT

She often works on the wrong thing because expectations were never clear from the start.

KEY NEED

Private onboarding and specific completion criteria.

Dito and Sinta are not lazy. Free-riding is more often driven by structure than motivation (Piezon & Ferree, 2008). SatuTim is designed for all three, not just Rahma.

SatuTim: AI that closes collaboration gaps

FOUR PHASES, ONE APP (WhatsApp is not the execution platform)

① MATCH
Upload CVs; IBM Bob extracts profiles and identifies skill gaps

② ASSIGN TASKS
AI breaks work into role-based subtasks, each with a Definition of Done (DoD)

③ MONITOR
Update status in one tap via a WhatsApp magic link

④ LOG
Contribution report and Collaboration Portfolio

A 7-step onboarding wizard; coordinator setup takes ≤5 minutes (assumption). Passive members can update their status with a one-tap WhatsApp magic link.

THREE EVIDENCE-BASED DIFFERENTIATORS

1 CV-based skill-gap analysis

Complementary skills improve output quality (van Knippenberg et al., 2004). SatuTim identifies gaps before work begins, not after.

2 Explicit roles and a DoD for each role

Role clarity is one of the strongest predictors of reduced social loafing (Oakley et al., 2004). AI generates a DoD for each task type instead of using a generic template.

3 Visible accountability, private notifications

Visible individual accountability reduces free-riding (Johnson & Johnson, 2009; Kerr & Bruun, 1983). Magic links keep friction to a minimum.

MVP TASK TYPES

T1	T2	T6
Paper / Report	Project / Prototype	Custom (manual)

T3–T5 are excluded from the MVP because Build Week lasts only three days.

COLLABORATION PORTFOLIO: THREE THEME 2 KEYWORDS

Collaborate: the product's core, fair task allocation, visibility, and accountability.

Connect: CV gap analysis matches members to roles based on their skills.

Engage with opportunities: a digital contribution portfolio can improve access to internships and jobs (Barrett, 2007).

Note: the adapted methods above (Kanban, DoD, async stand-ups) are exploratory, their effectiveness in Indonesian academic settings has not yet been tested.

IBM Bob throughout the workflow

21 DOCUMENTED SESSIONS, THREE MODES

Agent (sessions 01–23, used most often); Plan (sessions 03 and 23: assumption audits, option evaluation, and literature-task breakdown); Ask (session 12: good-practice guidance and best practices). Using multiple modes addresses a judging sub-criterion.

EXAMPLES OF PROMPTS USED

Agent (session 01): "Identify at least 8 real problems faced by Indonesian university students that relate to Theme 2..."

Agent (session 10): "Break the following task into role-based subtasks: [...]. Include a DoD and estimated hours."

Agent (session 10): "Calculate each member's workload percentage. Flag any difference above 25%."

OUTPUT QUALITY: TEAM REVIEW

Accepted: sessions 01, 03, 07–23. Accepted with notes: 02, 05. Partially accepted: 04. Revised: 06. Every output was reviewed before being integrated.

Build Week plan: IBM Bob will generate code for the CV-parsing, gap-analysis, and task-splitting endpoints. All sessions will be recorded in the LOG. (Planned, not yet completed)

HUMAN CORRECTIONS: EVIDENCE OF CRITICAL THINKING

CORRECTION #1 · SESSION 06 → 07

Bob recommended: Ideas 1+10, a Kickoff Pact and a WhatsApp-based AI Mediator.

The team rejected this because: it still depended on WhatsApp and skipped the Monitor and Log phases.

What changed: a unified four-phase workflow in one app; Section 5 was rewritten.

Impact: the concept shifted from a facilitator for two moments to a platform supporting the full workflow.

CORRECTION #2 · SESSION 08

Problem: plans/01–05 did not address task types or skill imbalance.

The team raised a real-world case: 20 students, 10 teams, 15 frontend developers and 5 backend developers. AI cannot recommend roles without gap analysis.

Added: Sections 5.2–5.3 in plans/05, forming the basis for F2 in the PRD.

23	3	12
documented sessions	Bob modes used	human corrections

Fairer collaboration, with a record to show for it

SHORT-TERM IMPACT: EVIDENCE-BASED MECHANISMS

Visible individual accountability is a key element of cooperative learning (Johnson & Johnson, 2009) and reduces free-riding when contributions are visible (Kerr & Bruun, 1983).

Rahma gets evidence of workload imbalance, not just a hunch. Dito and Sinta know their roles from the start, while one-tap magic links make status updates easier.

Visibility should provide feedback, not surveillance (Deci & Ryan, 2000). This is an assumption and has not been validated in the field.

LONG-TERM IMPACT

A digital contribution portfolio may improve access to internships and jobs, while showing more specific competencies than GPA alone (Barrett, 2007).

Students learn industry collaboration practices: Working Agreements, DoD, and async stand-ups. This is an exploratory adaptation; its transfer to academic settings has not been tested.

GUARDRAILS

> 50% of rooms stop at Phase 1: onboarding UX problem
> 30% report that notifications feel intrusive: design is too aggressive
< 20% of rooms produce a report: Phase 4 is too burdensome

PROPOSED SUCCESS METRICS

Targets

< 15%
contribution gap between the highest and lowest contributors in ≥ 60% of teams

≥ 80%
of members complete onboarding within the first 24 hours

≥ 70%
of subtasks are updated before the deadline

≥ 40%
of rooms produce a final contribution report

Until now, hard-working students have had no way to prove their contribution, while students who struggle have had no way to ask for help without embarrassment. SatuTim addresses both gaps, with AI that supports people, not monitors them.